

Name: _____, Student ID: _____

CSE 468 Fall 2026 Exam 2 Practice

Assignment ID: 49

Instructions

Don't forget to write your name and student ID at the top. **You must show your ID as you turn in your exam.** You have an hour and 15 minutes (a regular class period) to complete this exam (unless you are taking it in another location with different arrangements). Mark on these sheets of paper with a pen or pencil, and then turn it in at the front of the room to the TA or I. ~~You may not use scratch paper or notes of any kind, so there should be no other pieces of paper on your desk during the exam and you should not write on anything other than the exam itself.~~ **Update for Exam 2 only: The exam is open notes. You may have one sheet of paper with anything you want written on the front and back, but no other paper except this sheet and this exam should be on your desk.** This exam is closed book (note that there is no textbook for the course) and closed note. You may use a calculator, but only a Casio fx-115ES PLUS. You may not use any other electronic device (and especially not a cell phone). You may not communicate in any way with any individuals other than the instructor of the course, the TA, or another official proctor during the exam. Any violation of these policies will result in a 0 on the exam and will be treated as an act of academic dishonesty as per the syllabus. The exam is worth 100 points total. You must show your work for all questions. This is printed exam number: 49 (you can safely ignore this number).

Fermat's Little Theorem and fast modular exponentiation

(50 points) Calculate the modular multiplicative inverse for $e = 68$ for $GF(233)$, i.e. for arithmetic modulo 233, using Fermat's little theorem. You must show your work, and you must use fast modular exponentiation by repeated squaring (and show the work for that, too).

Multiplicative inverse is $68^{233-2} \pmod{233} = 24$

Note that on the exam you'll need to show your work. In particular, you need to show the steps of fast modular exponentiation *via* repeated squaring.

$$68^{-1} \equiv 68^{2^{33}-2} = 68^{2^{31}} \pmod{233}$$

$$231_{10} = 11100111_2$$

$$= 2^7 + 2^6 + 2^5 + 2^2 + 2^1 + 2^0$$

$$= 128 + 64 + 32 + 4 + 2 + 1$$

Repeated squaring:

$$68^2 = 4624 \equiv 197 \pmod{233}$$

$$68^4 \equiv 197^2 = 38809 \equiv 131 \pmod{233}$$

$$68^8 \equiv 131^2 = 17161 \equiv 152 \pmod{233}$$

$$68^{16} \equiv 152^2 = 23104 \equiv 37 \pmod{233}$$

$$68^{32} \equiv 37^2 = 1369 \equiv 204 \pmod{233}$$

$$68^{64} \equiv 204^2 = 41616 \equiv 142 \pmod{233}$$

$$68^{128} \equiv 142^2 = 20164 \equiv 126 \pmod{233}$$

Therefore:

$$68^{2^{31}} \equiv 126 \cdot 142 \cdot 204 \cdot 131 \cdot 197 \cdot 68$$

$$\equiv 184 \cdot 204 \cdot 131 \cdot 197 \cdot 68$$

$$\equiv 23 \cdot 131 \cdot 197 \cdot 68 \equiv 217 \cdot 197 \cdot 68$$

$$\equiv 110 \cdot 68 \equiv 24 \pmod{233}$$

Check: $68 \cdot 24 = 1632 = 7 \cdot 233 + 1$. So $68^{-1} \equiv 24 \pmod{233}$.

Diffie-Hellman (50 points)

Given prime $p = 337$ and primitive root $g = 166$, assume that you are Alice and your private key is 59. Bob sends you 290 over the network. What will you send to Bob over the network? (25/50 points)
After you've sent that to Bob, what will the shared secret key that you and Bob agreed on be? (25/50 points)

Answer to first part is 315, answer to second part is 310

Note that on the exam you'll need to show your work. In particular, you need to show the steps of fast modular exponentiation *via* repeated squaring.

To Bob: $A = g^a \bmod p = 166^{59} \bmod 337$.

$$59_{10} = 111011_2$$

$$= 2^5 + 2^4 + 2^3 + 2^1 + 2^0$$

$$= 32 + 16 + 8 + 2 + 1$$

Repeated squaring:

$$166^1 \equiv 166 \pmod{337}$$

$$166^2 = 27556 \equiv 259 \pmod{337}$$

$$166^4 \equiv 259^2 = 67081 \equiv 18 \pmod{337}$$

$$166^8 \equiv 18^2 = 324 \equiv 324 \pmod{337}$$

$$166^{16} \equiv 324^2 = 104976 \equiv 169 \pmod{337}$$

$$166^{32} \equiv 169^2 = 28561 \equiv 253 \pmod{337}$$

$$166^{59} \equiv 253 \cdot 169 \cdot 324 \cdot 259 \cdot 166$$

$$\equiv 295 \cdot 324 \cdot 259 \cdot 166 \equiv 209 \cdot 259 \cdot 166$$

$$\equiv 211 \cdot 166 \equiv 315 \pmod{337}$$

Therefore, I send 315 to Bob.

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Shared secret: $K = B^a \bmod p = 290^{59} \bmod 337$.

Using $59 = 32 + 16 + 8 + 2 + 1$ again:

Repeated squaring:

$$290^1 \equiv 290 \pmod{337}$$

$$290^2 = 84100 \equiv 187 \pmod{337}$$

$$290^4 \equiv 187^2 = 34969 \equiv 258 \pmod{337}$$

$$290^8 \equiv 258^2 = 66564 \equiv 175 \pmod{337}$$

$$290^{16} \equiv 175^2 = 30625 \equiv 295 \pmod{337}$$

$$290^{32} \equiv 295^2 = 87025 \equiv 79 \pmod{337}$$

$$290^{59} \equiv 79 \cdot 295 \cdot 175 \cdot 187 \cdot 290$$

$$\equiv 52 \cdot 175 \cdot 187 \cdot 290 \equiv 1 \cdot 187 \cdot 290$$

$$\equiv 187 \cdot 290 \equiv 310 \pmod{337}$$

Therefore, our shared secret key is 310.